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(1) Research Program/Project/Study Title: Maximizing Coconut Income through Toddy and Nut Production in a Single Spadix: A Modification to Sequential Toddy and Nut Production Technique

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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	
Province:	
Region:	

(5) Classification Research: ☐ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity

(8) Discipline/Field of Specialization

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Despite being the second-largest producer of coconuts in the world and the top exporter of coconut products, the Philippine coconut industry has been in a constant state of stagnation. The researcher was able to come up with another plan for producing toddy and nuts using the still underdeveloped coconut growing business. The study was conducted to determine the income to be earned from toddy and nut production in a single spadix utilizing modified techniques of sequential toddy and nut production in an effort to boost the struggling coconut agriculture sector. For the factorial experiment in this research, a completely randomized design was used. The yield of toddy per variety was shown to be strongly impacted by the section of the spadices after data collection and analysis. The middle portion of the spadix yielded more toddy. In comparison to the middle region of the spadix, the base section of the spadix had more buttons grow. The number of buttons and nuts gathered from the base of the spadix varies significantly among dwarf coconut cultivars. Section and variety had an interaction effect that affected how much toddy was produced. When the spadix was tapped at the tip and then allowed to develop nut, the highest income from toddy and nuts was attained.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Keywords: nuts, toddy, spadix, sequential, buttons

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews).

INTRODUCTION

Coconut (*Cocos nucifera* Linn) is one of the most important food and cash crops grown in the tropics. It is widely adapted to various environments and climatic conditions of many producing countries, occupying over 20% of available arable land (Magat, 1996). Coconut tree is known as the "tree of life" because of its several uses, including those in construction, agriculture, energy, and medicine (Feuersenger, 2012). It has provided humankind with many products and byproducts, making it essential to Philippine agriculture and the country's economy. It is considered as one of the largest employers of agricultural land and labor in the Philippines (Clarete and Roumasset, 1983). It is considered a predictor of the general economic activity of the country due to its performance as a significant source of income for the Philippines industry (PSA, 2022).

The Philippine coconut industry has been in a state of constant stagnation even though the country is the world's second-largest producer of coconuts and the top exporter of coconut products, boasting of its globally competitive coconut oil, virgin coconut oil, and desiccated coconut, among other products (DA Communications Group Article, 2022). Villegas (2021) made a statement in which he coined the Food and Fertilizer Technology Center for the Asian and Pacific Region (PCAARRD) and stated that the country's coconut agriculture is still regarded as an orphan due to inadequate investment, with a very high incidence of poverty among millions of Filipino coconut farmers. Because of the low and unstable price of copra, it is becoming more and more difficult for small coconut farmers in the Philippines to rely primarily on its output. Production methods that boost output, revenue, and farm sustainability are required and have to be relatively simple for

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farmers to embrace (Maravilla & Magat, 1993). With this, there is a need to develop a diversified coconut production scheme to be adapted by the coconut farmers.

One way of doing this is by producing toddy from the cut unopened inflorescence of the tree, known as tapping, while producing nut out of the untapped portion of the spadix. Tapping is almost a culture among most Filipinos, especially those living in the Visayas and Mindanao Islands. It could be advertised as locally popular fresh adulterated beverages. Tuba, in the form of vinegar (under natural fermentation), and lambanog, or gin, as a source of lactic acid. Additionally, syrup, unrefined sugar, or crystallized sugar can be made from the sap (Fremond, 1966). Tagarino (1993) reported that on a per-hectare basis, the production of tuba and nuts on a farm is more profitable than on farms producing nuts alone.

Even though Toddy production is an important component of the Philippine coconut industry, there has been little research. This explains why existing tapping technologies have all been developed by tappers themselves, and few have come from the scientific community (Santiago and Virtudazo, 1996).

In the researcher's earlier research, a scheme was devised that divided the spadix into two sections for the production of toddy and nuts. Still, the farmers found it difficult to adopt this method because it only produced a small amount of toddy. However, this approach results in more nuts.

With the still unprogressive coconut farming industry, the researcher was able to devise another scheme for toddy and nut production. This time, the spadix was divided into three sections. In an effort to support the suffering coconut agriculture sector, this study was started to determine whether this scheme will increase toddy and nut yields.

OBJECTIVES OF THE STUDY

The study was conducted to determine the income to be generated from toddy and nut production in a single spadix using a modified techniques of sequential toddy and nut production: It specifically aimed to:

1. determine the amount of toddy produced per section of the spadix;
2. quantify the number of the nuts produced per section after tapping;
3. compute the income derived per section when toddy and nut were produced; and
4. determine the interaction effect between the sections of spadix and different dwarf coconut varieties in terms of toddy yield.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

MATERIALS AND METHOD

The spadix of the coconut was divided into three (3) sections; the tip, middle, and base sections. The spadix tapped until it reached the middle portion of the spadix, likewise on the base section of the spadix. The study was conducted using the Completely Randomized Design for the factorial experiment. There were two factors considered, Factor A- Varieties and Factor B –Section of spadices. Varieties used were V1 – Aromatic (ARO), V2 – Equatorial Green Dwarf (EGD), V3 – Sri Lanka Green Dwarf (SGD), V4 – Talisay (TAL), V5 - Malayan Yellow Dwarf (MYD), V6 - Malayan Red Dwarf (MRD) and V7 - Pilipog (PIL).

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Section of spadix included S1- Tip section, S2 – Middle section, and S3- Base section. Each treatment was replicated three times with six trees per treatment. All data gathered were subjected to the Analysis of Variance using LSD test and interpreted at and 5% levels of significance. DMRT was used for the interaction effect.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

RESULTS AND DISCUSSION

This chapter showed the results that were gathered by the researchers and its discussions.

Volume of Toddy per Section (ml)

Figure 1 showed that among the varieties utilized in the study, SGD variety (2,502.24 ml) produced greater yield of toddy from the middle section of tapped spadix during dry season which is significantly differed from the other varieties and section of the spadix utilized in the study. While no toddy was produce from the base section Talisay variety was noted.

In general, more toddy was produced from the middle section of the spadices while a comparable least yield of toddy was produced from tip and base section of the spadices. Among the variety, SGD produced the greater amount of toddy which is found to be comparable with the toddy produce from ARO and EGD. While lesser toddy produced from MYD variety (1086.50) ml and found to be comparable with MRD, TAL, PIL. There are some factors to consider in tapping our plant, some are age of the plant, health of the palm, genetic make-up, size and age of the spadix, and morphological traits of the palm are plant factors known to affect toddy yield. On the other hand, some of the external factors that affect toddy yield are climatic conditions, soil fertility, and tapping practices (e.g., duration of tapping, number of spathes tapped, and skill of the tapper).

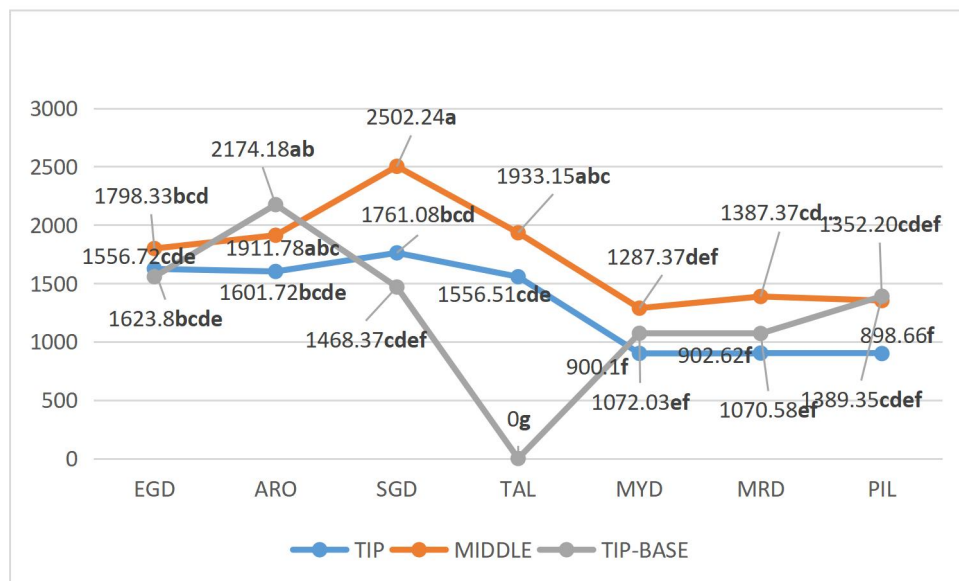


Figure 1. Volume of toddy per section of the spadices tapped during dry season.

Number of Buttons Left and Nuts Harvested

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Analysis of variance in Figure 2 revealed that the middle portion of the spadix did not influence the number of buttons left after tapping likewise the number of matured nuts harvested of different dwarf coconut varieties.

Also, it was noted that the MRD variety produces the most buttons (4.22) and nuts (2.97) after tapping. This was followed by a PIL with 3.66 numbers of buttons and 2.11 numbers of nuts produced, then the EGD variety with 3.39 buttons and 1.89 nuts produced and SGD with 2.98 buttons and 1.83 nuts produced. In contrast, the ARO variety produced the least number of buttons (1.67) and nuts (1.11) among the different dwarf varieties. Figure 3 also showed us that TAL and MYD produced the same amount of nuts (1.78), but differed in the production of buttons (2.67 and 3.33, respectively).

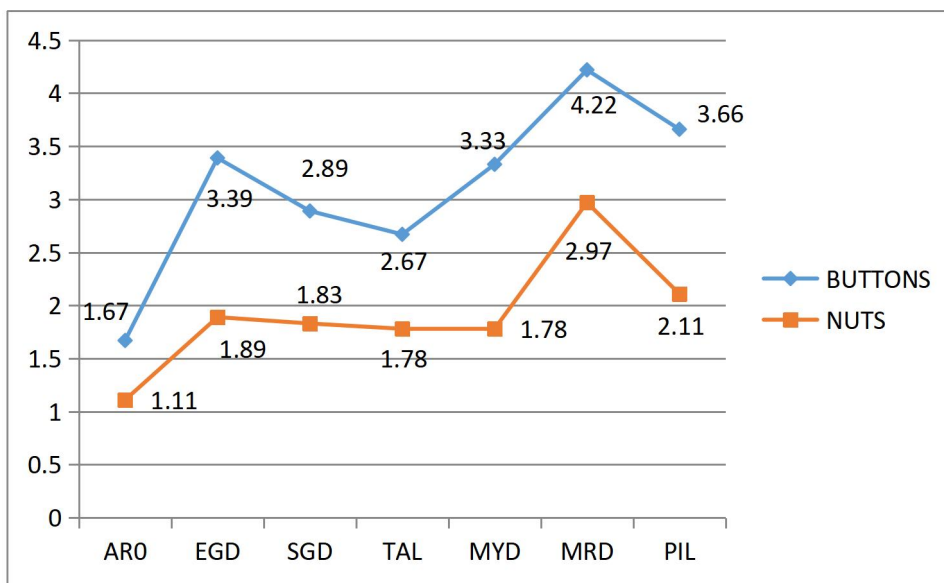


Figure 2. Number of buttons left and nuts harvested from the middle portion of tapped spadix of different dwarf coconut varieties.

The result of ANOVA revealed that the different dwarf variety of coconut significantly differed in producing buttons and nuts after tapping in the base portion of the spadix. For the number of buttons left, Talisay variety had the most number of buttons left after tapping with the mean of 7.78 nuts which is significantly different from that of EGD (6.17), ARO (4.5) and Pilipog (3.89). Least number of buttons was observed from variety of SGD (3.5) which is found to be comparable with Pilipog (3.89) and MYD (3.56).

The results further revealed that most number of the harvested coconut was recorded from EGD which is found to be comparable from that of Talisay (2.78), SGD (2.66), Pilipog (2.56) and MRD (2.28).

Dela Cruz (2009) examined the various dwarf and hybrid coconut cultivars based on the properties of the nuts they produce. The MRD is one of these variants. As for the nut's shape, an oval shape has been seen in this kind. Additionally, the lightest nut weight, thickest shell, lightest shell weight, and lightest meat weight were all noted. MRD variant nuts are reported to be thicker.

On the other hand, similar study was conducted by Legaspi, et.al in 2018 which one of the variety characterized was the Talisay. They reported that Talisay has a comparable heavy shell weight and a thicker meat among the different dwarf varieties characterized. It also possessed an angled fruit shape and a round nut shape.

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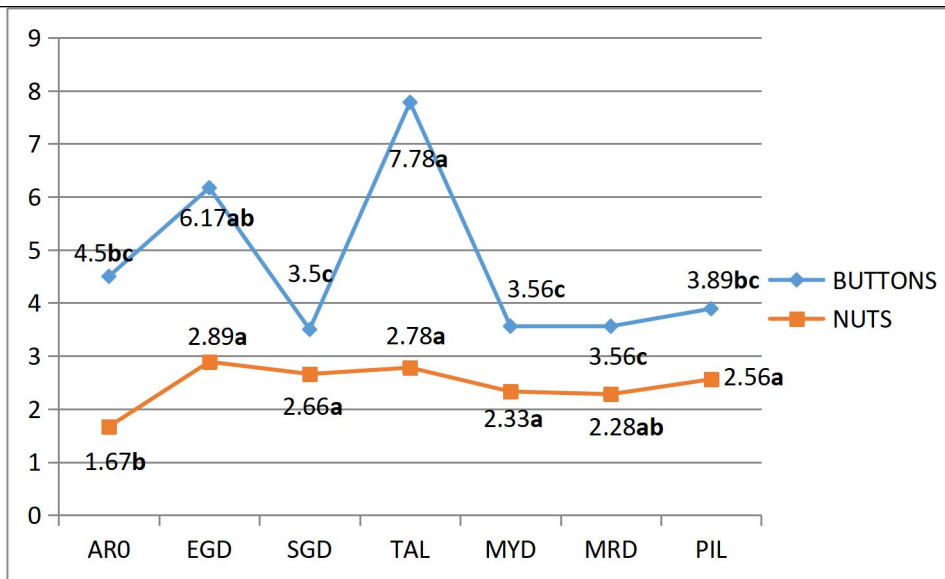


Figure 3. Number of buttons left and nuts harvested from the base portion of the tapped spadix of different dwarf coconut varieties.

General observations

It was noted from the result of the study that more buttons were left after tapping at both the tip and based sections however, almost 50% of them do not mature but were dehisce. This could be due to the health status of the coconut tree, environmental condition such as temperature and moisture (Samaranghe, et.al. 2018). Another possible contributing to low fruit setting in tapped spadix is the stress impact of tapping although this has to be investigated.

Income Generated from Toddy Production Tapped up to Different Portions

Figure 4 displayed the income generated from toddy production tapped up to the different portions of the dwarf coconut varieties.

Based on the findings, it can be noted that when the spadix was tapped up to the middle portion, five (5) of the dwarf varieties produced high income in terms of toddy production, with SGD earning the income of 100.09, followed by TAL (77.33), ARO (71.93), MRD 55.49), and MYD (51.49). When the spadix was tapped up to the base, only two (2) of the types showed substantial revenue creation: EGD and PIL, each with an income generation of 86.97 and 55.57 respectively. In contrast, the spadix that was tapped up to the tip produced the least amount of toddy, with MRD (36.10), MYD (36.00), and PIL (35.95) generating the three lowest amounts of revenue, followed by TAL, which produced no toddy when tapped up to the base.

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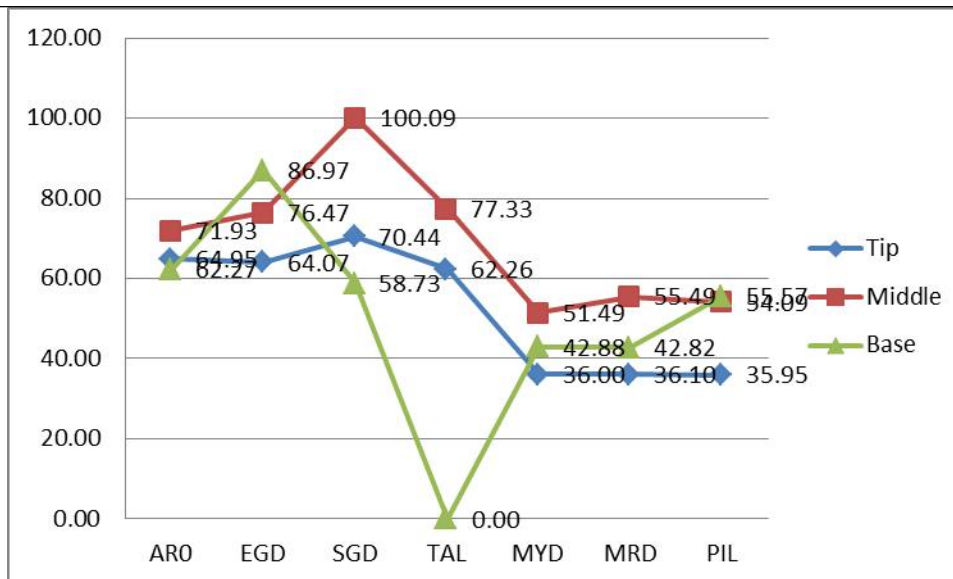


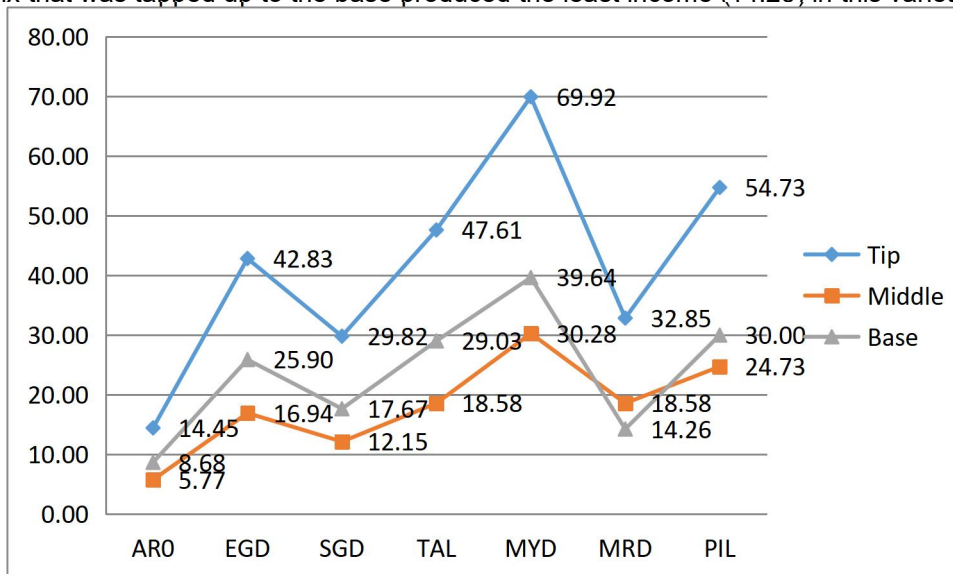
Figure 4. Income generated from toddy production tapped up to different portions from different dwarf coconut varieties.

Income Generated from Nut Production Tapped up to Different Portions

Figure 5 revealed the income generated from nut production tapped up to different portions from different dwarf coconut varieties.

The results shown in the figure indicate that the highest income from nut production was generated by all coconut varieties when the spadix was tapped up to the tip portion. The highest revenue will come from MYD, with 69.92, followed by PIL (54.73), TAL (47.61), EGD (42.83), MRD (32.85), SGD (29.82), and ARO (14.45). This means that to maximize profit in the nut production, it is best to tap the spadix up to the tip portion.

In contrast, it was observed that the spadix that was tapped up to the middle portion of all the varieties, including MYD (30.28), PIL (24.73), TAL (18.58), EGD (16.94), SGD (12.15), and ARO (5.77), produced the least income, with the exception of the MRD Variety, where it was noted that the spadix that was tapped up to the base produced the least income (14.26) in this variety.



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Figure 5. Income generated from nut production tapped up to different portions from different dwarf coconut varieties.

Total Income Generated from Nut and Toddy Production Tapped up to Different Portions

Figure 8 revealed Total income generated from nut and todody production tapped up to different portions from different dwarf coconut varieties.

When the revenue from todody and nut production is added up, it is found that EGD tapped up to the base will produce the most revenue among all other varieties, with an income totalled to 112.86. While TAL found that the lowest income was produced when Spadix was tapped up to base, with an income of 29.03.

The varieties SGD (112.24) and MRD (74.08) showed their highest income generation when the spadix was tapped up to the middle, while the varieties TAL (109.87), MYD (105.93), PIL (90.68), and ARO (79.40) showed their highest income generation when the spadix was tapped up to the tip.

Contrastingly, when the spadix was tapped up to the base, five (5) of the varieties—PIL (85.58), SGD (76.40), ARO (70.95), MRD (57.09), and TAL (29.03)—showed its lowest revenue generation. While when spadix was tapped up to the middle, the varieties EGD (93.41) and MYD (81.78) displayed their lowest income generation.

To maximize profit it is suggested to use EGD variety and have the spadix to be tapped up to its base.

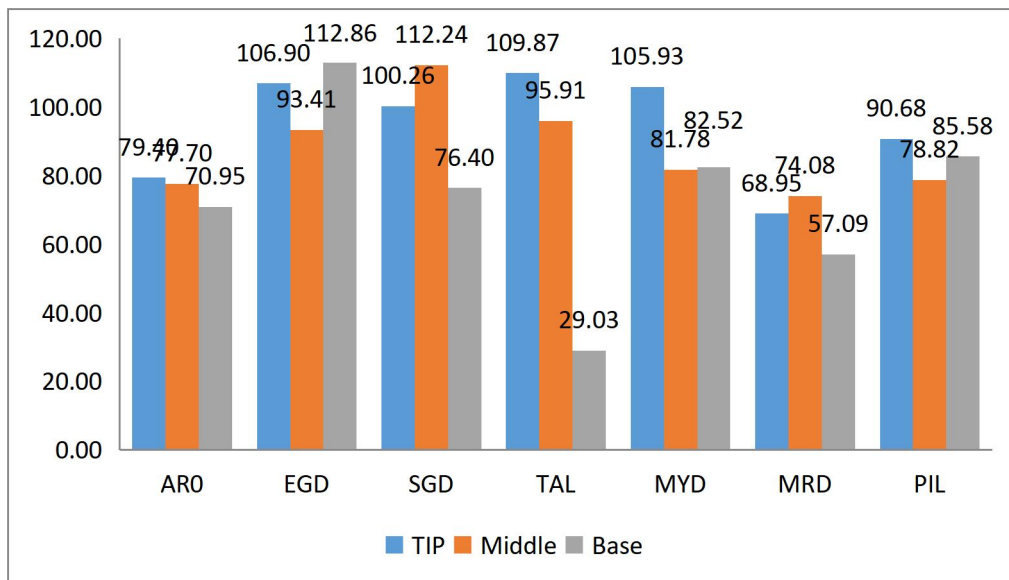


Figure 8. Total income generated from nut and todody production tapped up to different portions from different dwarf coconut varieties.

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSION AND RECOMMENDATIONS

Based on the results of the study the following conclusions were drawn:

1. Section of the spadices significantly affected the yield of toddy per variety.
2. Greater volume of toddy was produced in the middle section of the spadix.
3. SGD manifested the highest volume of the toddy produced in middle section.
4. MRD produced the mst number of buttons and nuts harvested.
5. More buttons developed in the base section of the spadix compared to the buttons developed in the middle section of the spadix.
6. Different dwarf varieties of coconut have a significant influence in the number of buttons and nuts harvested from the base section of the spadix.
7. There was an interaction effect of section and variety in influencing the yield of the toddy.
8. Among the dwarf varieties when tapped at a certain section and allowed to bear fruit after, TAL gave the highest income when tapped at the tip, SGD when tapped until the middle portion, and EGD when tapped up to the base portion.
9. For combined income from toddy and nut, highest income was obtained when spadix was tapped at the tip and allowed to produced nut after.

Based on the results of the study, the following recommendations are forwarded:

1. For greater volume toddy tapped the spadix from the tip to middle section and let the base section to bear fruit.
2. Use SGD variety for greater volume of toddy.
3. Different dwarf varieties can be utilized in the production of toddy at the same time nut in a single tree.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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